

Zohair Omar

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Role-Specific Portfolio: zohairomar.netlify.app/for/gdms/software-development-test

Summary

4th Year Computer Science & Mathematics student @ UofC (transferred from UBC) with a Minor in Economics; **GPA: 3.4**. Hands-on across the languages this role calls for: shipped a Java/Spring Boot REST API with **99 unit and integration tests** through Docker-based CI/CD; built a Python/Flask RAG service integrating OpenAI and Gemini with multi-provider fallback; built embedded **Arduino/C-based path planning and robot control** on a UBC Design League autonomous robot (95% navigation success rate). Comfortable across **C, C++, Python, Java, JavaScript/TypeScript, and SQL**; testing-and-debugging discipline reinforced by automated CI test gates and code review on every merge. **Based in Calgary, eligible to work in Canada**, returning to full-time studies after the term; **prefer the 12-month term** for long-term integration with the team.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, R, HTML, CSS

Embedded & Real-Time: Arduino C, microcontroller-based robot control, servo and ultrasonic sensor integration, real-time path planning, SolidWorks 3D modeling, rigid body force analysis

Testing & QA: JUnit5, pytest, unit and integration testing, automated CI test gates, regression testing, code review, debugging, integration verification

AI/ML: OpenAI API, Google Gemini, RAG systems, prompt engineering, multi-provider fallback, full-stack AI applications

Backend & Data: Spring Boot, REST APIs, Flask, OAuth 2.0, RBAC, PostgreSQL, MySQL, SQL Server, Pandas, statsmodels

DevOps & SDLC: Git, GitHub Actions, Jenkins, Docker, CI/CD, release management, Agile/Scrum, sprint planning, Linux

Education

Bachelor of Science, Computer Science & Mathematics (Minor in Economics) Expected Apr 2027
University of Calgary, Alberta

Bachelor of Science, Computer Science Sep 2022 – Sep 2024
University of British Columbia, Vancouver

Relevant Coursework: Software Engineering, Object-Oriented Programming (UML), Data Structures & Algorithms, Operating Systems, Applied Machine Learning (Python), Statistics, Functional Programming

Projects

Incident Management Platform; Java, **Spring Boot**, PostgreSQL, GitHub Actions, Jenkins, Docker

- Delivered a full-stack **Java/Spring Boot REST API** backend with **object-oriented design**, UML class diagrams, ERD-based PostgreSQL schema, and role-based access control across ticket, incident, and release workflows.
- Shipped **99 unit and integration tests** validating every merge through a CI/CD pipeline using GitHub Actions, Jenkins, and Docker; performed integration testing, debugging, and code review across feature branches.

UBC Design League – Autonomous Robotics Project; C, Arduino IDE, Python, SolidWorks, Git Dec 2022 – Sep 2024

- Built an **Arduino-based path planning and robot control system in C** with servo motor and ultrasonic sensor integration; **95% navigation success rate** on object detection tasks.
- Performed rigid body force analysis and travel calculations; 3D modeled and prototyped components in SolidWorks; improved real-time perception efficiency by **25%** through sensor fusion tuning.

Ticket Triage RAG Bot; Python, Flask, OpenAI API, Google Gemini

- Built a Python/Flask REST service integrating OpenAI and Gemini APIs with multi-provider fallback, structured prompt engineering, caching, and **34 automated tests**; AI/ML system using retrieval-augmented generation patterns.

Experience

Enactus UCalgary – WealthPath Jul 2025 – Present
Front-End Development Lead; React, TypeScript, Google Cloud MySQL, GitHub Actions Calgary, AB

- Led a 4-person Agile team through weekly sprints; built a reusable React/TypeScript component library and connected the frontend to a Google Cloud MySQL backend; ran **30+** merges through CI/CD with code review.

TechNova Feb 2025 – Present
Co-Founder & VP Technical; Python, SQL, Git Calgary, AB

- Co-founded a technology consulting club; built full-stack web applications and dashboards for **100+ users**; managed 4 developers through Agile sprints across concurrent client engagements.

Agrius IT May 2024 – Jul 2024
Software Developer Intern; Python, SQL, HTML/CSS, JavaScript Abu Dhabi, UAE

- Developed and debugged **Python/SQL backend** solutions for an enterprise SaaS platform serving **1,000+ daily users**; implemented CORS and RBAC; resolved three critical bottlenecks ahead of schedule.